From Clocks to Coordinates: Controlled Human Perturbations Reveal Hidden Directions in ECG Biological Age

A condensed account of the central result

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Abstract

Neural "ECG age" and organ-age models return a single scalar, collapsing biophysically distinct processes into one number. This work shows biological age is better read as a *direction* in a state space. Four phase-of-heartbeat age clocks compress into a *shared axis* A and an orthogonal *disagreement radius* D, with the geometry frozen on CODE-15 *before any outcome was read*. The radius D was null on its single pre-registered confirmatory mortality test (hazard ratio 1.01 per SD, 95% CI 0.98 to 1.04, $p = 0.48$)—the honest negative that motivates a signed analysis. A randomized I_{Kr} blocker (dofetilide) displaced the ST-T repolarization clock (+4.2 SD·h, false-discovery-rate $p = 0.002$), defining a covariance-scaled signed *direction* $\mathbf{w}_{I_{\mathrm{Kr}}}$—the coordinate the unsigned radius discards. It was confirmed by a second I_{Kr} blocker (quinidine, +0.83) and was specific relative to non-I_{Kr} comparators; direction-derivation is exploratory (Holm-adjusted permutation $p = 0.069$). Frozen and carried into an independent external cohort (Chapman–Shaoxing/Ningbo, $n = 44\,550$, 386 QT-extension cases), it added *conditional* information about physician-assigned QT-interval extension beyond conventional intervals, A and D (fully adjusted odds ratio 1.23 per SD, $p = 1.9 \times 10^{-4}$; marginally null—a suppression effect). The shared axis A was separately mortality-associated and generalized across the body. This condensed account presents that result; a 53-page technical manuscript provides the full methods, the multiscale atlas (whole-body CT, brain MRI, an *in-silico* genomic screen), and every frozen artifact.

1 The claim

Biological-age models—whether a neural network reading an electrocardiogram or a volumetric organ clock—almost always return *one number*: an age gap. That scalar discards direction. Two people equally "old for their age" may be old in physiologically opposite ways. The question this work answers is whether that lost direction is real, measurable and *transportable*: can a controlled human perturbation reveal a fixed coordinate in an age–state space, and does that coordinate carry information in a cohort that never saw the perturbation?

2 A shared axis that behaves like aging, and a radius that does not

The shared axis A behaves the way a biological-age summary should. In frozen CODE-15 development it was associated with mortality (hazard ratio 1.22 per SD); it generalized to whole-body organ-system aging in NHANES (+1.21 per SD, 95% CI 1.13 to 1.29) and transferred geographically to a Chinese cohort ($r = 0.65$; Fig. 2).

One heartbeat, four electrical phase-clocks: a shared aging axis A and a disagreement radius D

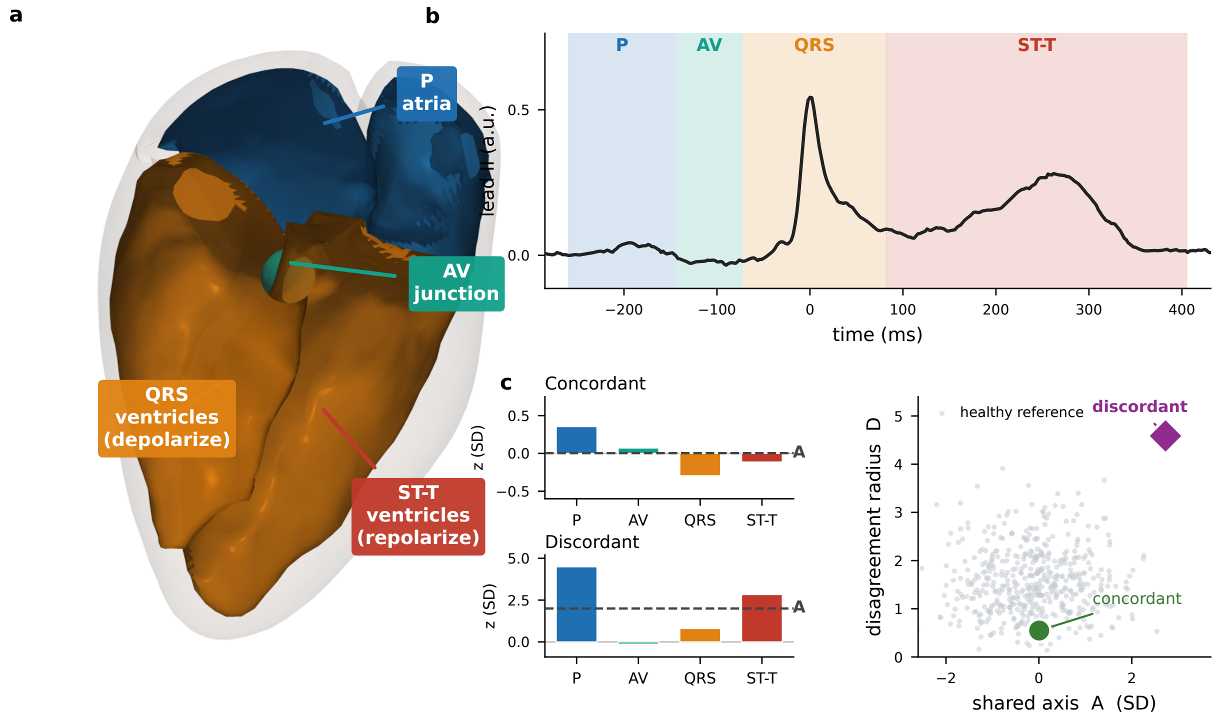

Figure 1: Biological age as a vector, not a scalar. Four phase-of-heartbeat age clocks (P, AV, QRS, ST-T) are standardized against a frozen healthy calibration cohort and read as a point in a four-dimensional gap space. Its coordinates are a *shared axis* A (the common displacement—how uniformly old the heartbeat looks) and an orthogonal *disagreement radius* D (how much the four clocks disagree). The geometry was frozen on CODE-15 (233 770 patients) before any outcome was read.

The disagreement radius D was the original hypothesis: that *how* the clocks disagree adds prognostic information beyond the shared level. It does not. On its single pre-registered confirmatory test—a model-frozen Cox likelihood-ratio test on the locked CODE-15 final-test partition—D was null (hazard ratio 1.01 per SD, 95% CI 0.98 to 1.04, $p = 0.48$; $n = 74\,715$, 2583 deaths). It showed no detectable association with independent cardiac structure and did not change under acute physiological stress (Fig. 3). Reporting that null honestly is what motivates the rest of the paper: the *unsigned* radius throws away a sign, and the sign is where the mechanism lives.

3 A controlled human perturbation reveals a hidden direction

The decisive experiment is a randomized-controlled human ion-channel perturbation. In the ECG-RDVQ crossover study, a specific I_{Kr} (rapid delayed-rectifier potassium current) blocker, dofetilide, was administered against placebo. I_{Kr} blockade has a textbook electrophysiological signature: it prolongs repolarization, that is, the ST-T segment. If the age–state space is physically meaningful, this perturbation should displace the ST-T phase clock along a *reproducible direction*.

It does. Dofetilide displaced the ST-T repolarization clock by +4.2 SD·h (false-discovery-rate $p = 0.002$), while a negative-control QRS contrast was null. From the mean placebo-corrected displacement $\mathbf{m}$, a covariance-scaled signed direction was defined,

$$\mathbf{w}_{I_{\mathrm{Kr}}} = \Sigma_q^{-1}\mathbf{m}/\sqrt{\mathbf{m}^\top \Sigma_q^{-1}\mathbf{m}},$$

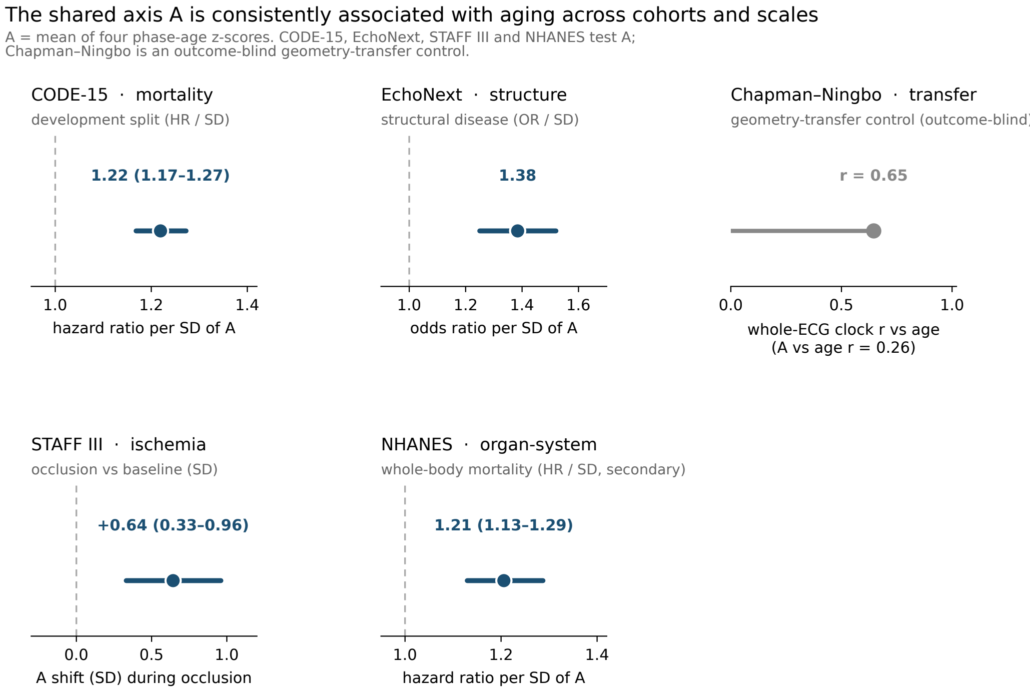

Figure 2: The shared axis A is consistently associated with aging across cohorts and scales. Development mortality, external geographic transfer, and whole-body organ-system aging (NHANES) all point the same way. A is the part of the decomposition that behaves like a conventional biological-age summary.

precisely the coordinate the unsigned radius D discards. Two independent checks make it more than a within-study artifact. A *second*, chemically distinct I_{Kr} blocker (quinidine) projected positively on the frozen axis (+0.83, 95% CI 0.52 to 1.20), a held-mechanism confirmation. And the direction was specific: non-I_{Kr} comparators (ranolazine, verapamil) had projections whose confidence intervals include zero (Fig. 4b). Because two perturbation directions (I_{Kr} and acute ischemia) were screened, the honest multiplicity-corrected permutation test on the direction's magnitude is only nominal (Holm-adjusted $p = 0.069$); the direction-*derivation* is therefore reported as exploratory, and its confirmation comes from the external cohort below.

4 The direction transports to an external cohort

A direction is only useful if it carries information where it was never trained. The frozen $\mathbf{w}_{I_{\mathrm{Kr}}}$ was carried, unchanged, into an independent external cohort (Chapman–Shaoxing and Ningbo, $n = 44\,550$; 386 patients with a physician-assigned diagnosis of QT-interval extension beyond conventional intervals). Added to a model already containing the shared axis A, the disagreement radius D and the conventional intervals, the signed I_{Kr} score carried *conditional* information about that diagnosis (fully adjusted odds ratio 1.23 per SD, 95% CI 1.10 to 1.36, $p = 1.9 \times 10^{-4}$; Firth-penalized odds ratio 1.22, $p = 1.9 \times 10^{-4}$). The marginal association was null (odds ratio 1.00)—a genuine suppression effect: the direction speaks only once the shared displacement and the conventional intervals are held fixed, which is exactly what a hidden *orthogonal* coordinate should do. The per-site estimates agree (Ningbo odds ratio 1.20, $p = 0.0019$, 329 cases; Chapman–Shaoxing odds ratio 1.44, labeled underpowered at 57 cases).

5 Why this is the result, and what the rest of the paper is

The claim is narrow and falsifiable: a controlled human perturbation of a single ion current defines a fixed, covariance-scaled direction in an ECG age–state space; that direction is the signed coordinate an unsigned age-gap radius discards; and, frozen, it transports to an independent cohort as conditional information about a clinically assigned repolarization phenotype. The honest negative (D is null on its confirmatory test) and the honest hedge (direction-derivation is exploratory under multiplicity; the external cohort provides the confirmation) are part of the result, not footnotes to it.

The full 53-page manuscript carries the complete methods, the frozen-geometry protocol and firewall, the acute-ischemia analysis (which moves A, not a stable signed contrast, and explains why ischemia fails the direction gate), and a deliberately *supporting* multiscale atlas—whole-body CT, brain MRI and an *in-silico* genomic screen of Human Accelerated Regions—that illustrates the same shared-axis/disagreement decomposition at other scales without claiming to confirm the electrical result. Those arms are the visually rich companion atlas; the perturbation direction above is the finding.

Aging is not one number. It is a state space—and physiology moves through it in directions.

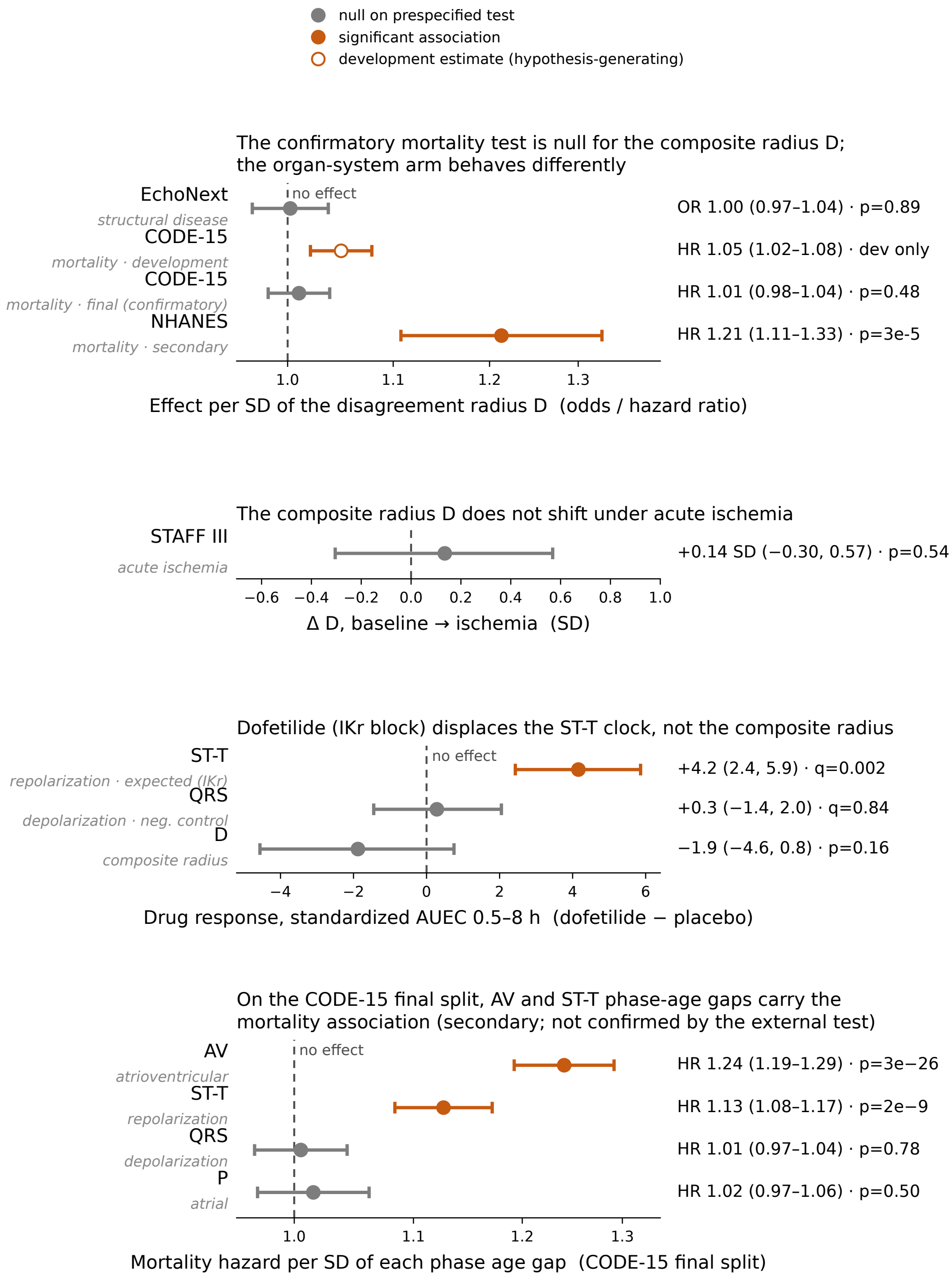

Figure 3: The unsigned radius D shows no detectable association with structure or response to acute stress, and is null on its confirmatory mortality test. The original radial hypothesis fails. The signed coordinate that the radius discards, however, is not empty—it is the subject of Fig. 4.

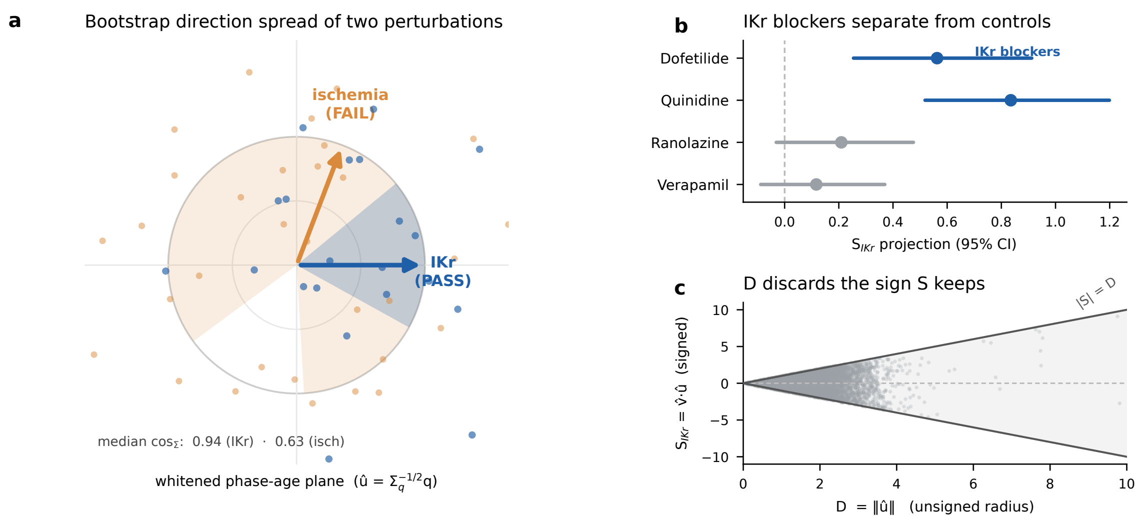

Figure 4: An outcome-blind, signed perturbation direction recovers a mechanism the unsigned radius discards, and is frozen for external transport. **(a)** Bootstrap spread of the recovered direction in a two-dimensional slice of the three-dimensional whitened contrast space: the I_{Kr} axis (blue) is tightly concentrated (bootstrap median $\cos_\Sigma = 0.94$; 9999 of 10 000 bootstraps reproduce its sign with no alignment), whereas the acute-ischemia axis (orange) is diffuse because ischemia moves the vector along the shared axis A rather than within the contrast subspace. **(b)** The two I_{Kr} blockers separate from zero; non-I_{Kr} comparators do not. **(c)** In raw whitened coordinates the signed score is bounded by the unsigned radius, $|S_{\mathrm{raw}}| \leq D_{\mathrm{raw}}$: the radius keeps the magnitude and discards exactly the sign the perturbation axis recovers.